

A/B/G/P Final-Lock Review V4

Metalogic Labs | 18 September 2026

Status: REVIEW_PENDING | IMPLEMENTATION_QUALIFIED | NOT_FROZEN

Confirmation: ABGP-CONFIRM-v1 untouched

This is the final pre-freeze review package for the A/B/G/P study. It supersedes V3 only for current readiness. V3 remains unchanged as the historical record of the earlier implementation and power objections.

No confirmatory result is reported here. ABGP-CONFIRM-v1 has not been used. This document does not itself authorize freezing or confirmation.

1. Exact candidate identity

```
RealityGraph branch: abgp-preregistration-freeze-v1
Implementation head: bed8a4e5d35e211e7bdebca42f39d923f264fb70
Candidate digest: 771793f52cc8228c54bf2ac8ab46ebbc0c7a99284d94e3042951a7976921e04
Implementation qualification digest:
a7a9e40a8c55427ae35e475fb3f50f3a3f4c06e00db79c69d5aa8e57f866cfd3
Planning proposal digest: a38c94cf8180984f671936158182e72ebb2523e0c85a22211dd9b56671df9f34
Implementation qualified: true
Complete-PASS power qualified: false, pending joint planning approval
Freeze authorized: false
Confirmatory namespace used: false
```

The final candidate binds the design and analysis plans, executed A/B/G/P generators, registered analysis and validity logic, exchangeability code, P acquisition/invocation/posterior workers, implementation qualification, planning review, and the final-candidate builder itself by SHA-256.

Exact-head CI is green: DEV preregistration/full regression run 35285093632; executed boundaries/final-lock candidate run 35285093739; boundary workflow full regression 280 tests, all OK.

2. Resolution of the three remaining methodological review points

A/P - inferential unit and earliest generated ancestry

The current executed matrix makes the inferential unit explicit for every arm and audits generated roots at that unit. A uses an acquisition-to-sealed-future episode. P uses acquisition -> hard restart -> four nested source-distinct future probes collapsed to one episode outcome. B uses an ordered-direction latent-world pair with fresh latent world plus independently generated source and target grammar roots. G uses the world; doses are repeated measurements nested within it.

The ancestry audit establishes the declared generated dependency graph and reports no shared generated ancestor across scored units. Fixed protocol objects are shared by design. It deliberately does not claim that this audit alone proves statistical independence against arbitrary untracked global state.

B - IUT, grammar independence and ordinary-explanation control

B now executes all 12 ordered directions across four independently parameterized grammar families: extensional, compositional, reachability and constraint/order. Source and target grammars use disjoint surface symbols and primitives, different primitive counts/arity patterns, different serialization schemas, different inference routes and independent grammar seeds. No deterministic cross-grammar translation or one-to-one primitive dictionary is supplied.

Acquisition learns a behavioral equality-pattern capability from the source representation and protected source order, then applies that capability to the target grammar under all four preregistered structural interventions. Wrong-class, shuffled-coupling and old-information/bisimulation controls are executed from their permitted information.

The statistical construction is unchanged: 12 directions x 3 primary controls = 36 required components. The B arm p-value is the maximum of those 36 p-values because PASS requires all 36 alternatives. The 0.0125 planning alpha is the conservative first Holm threshold across the four arm-level p-values, not an additional 36-way Bonferroni correction inside B.

G - world-level exchangeability

G now uses an executed paired-cell structural world and a cell-dependent protected evaluator. Relevance is computed before corruption by intervention against that evaluator. Corruption count and magnitude are matched within each pair and the full dose

schedule is fixed with no adaptive stopping.

For each world the generator has one fair binary role assignment: left-side relevant or right-side relevant. The exact two-assignment potential-outcome law is enumerated under the registered sharp null and the complete within-world paired outcome vector must be invariant under the joint relevance-label exchange. The randomization block is the world, not individual cells or doses. This is scoped to the frozen finite paired-cell generator.

3. P persistence boundary now executed

P is no longer a same-process serialization demonstration. Invocation runs in a fresh restricted Python worker image with the retained structural object as the only experimental state supplied to the invocation path. The invocation image contains no acquisition/search routine. A kernel seccomp filter is installed before experiment input is read and the implementation probes forbidden file, network, process and inherited-descriptor access.

Executed P gates require source-distinct future carrier and serialization; label-free applicability; zero verifier calls after restart; zero pre-action reconstruction/search; targeted lineage deletion; explicit reacquisition through the frozen acquisition procedure; and restored performance after reacquisition. The current P implementation gate is true. This is implementation/mechanism qualification only, not a confirmatory P result.

4. Power issue identified in V3 and proposed resolution

V3 correctly showed that using a true planning effect equal to the observed PASS floor leaves the significance-plus-observed-floor event near 0.5. The scientific PASS floors are therefore kept unchanged while the planning alternatives are proposed strictly above them. The proposed counts are unchanged from the reviewed candidate counts.

Arm	Proposed count	Observed PASS floor	Planning alternative	Conservative lower bound
A	4,096 episodes	5 pp	10 pp	0.998000
B	1,015/direction = 12,180	15 pp	25 pp	0.980978
G	421 worlds	15 pp	25 pp	0.980495
P	4,096 episodes	5 pp	10 pp	0.995334

These are conservative planning calculations, not predicted confirmatory outcomes.

For B, the proposal additionally assumes a 0.95 true all-four-intervention world agreement rate when planning power for the observed >=0.90 pooled agreement gate. At 12,180 independent world units, the conservative agreement-gate power rounds to 1.0 in the candidate calculation.

For P, deletion-to-cold closeness is proposed as an enforced mechanism condition rather than a separate stochastic advantage. The deletion operation removes the retained lineage; both deleted and cold invocation paths then receive the same null retained state. Reacquisition must re-enter the frozen acquisition procedure and restore the pre-deletion capability.

5. What is being requested for joint approval

No new scientific hypothesis, observed PASS floor, Holm/IUT rule, source-distinctness requirement, inferential-unit definition or confirmatory result is being introduced here.

- 1. **Planning alternatives:** A = 0.10, B = 0.25, G = 0.25, P = 0.10.
- 2. **Counts:** A = 4,096; B = 1,015 per ordered direction; G = 421; P = 4,096.
- 3. **B planning agreement:** 0.95 all-four-intervention world agreement for planning the observed >=0.90 gate.
- 4. **P deletion condition:** deletion-to-cold equality is treated as a mechanically enforced gate rather than a separately powered effect.
- 5. **Text-implementation consistency:** the implementation summarized above matches the jointly specified A/B/G/P study closely enough to freeze the exact normative package.

APPROVED AS WRITTEN

Any revision should identify the item and replacement value or wording before freeze.

6. Freeze rule after approval

Approval of this review does not itself execute confirmation. After explicit joint approval, and only then: (1) update the normative planning text to the exact agreed wording; (2) hash the resulting exact repository tree and required implementation/environment artifacts; (3) mark the lock FROZEN; (4) verify the confirmatory namespace remains unused at the frozen tree; and (5) enable ABGP-CONFIRM-v1 for one confirmatory execution.

Any post-freeze scientific change requires a new preregistration/version and a fresh confirmatory namespace.

7. Interpretation boundary

implementation_qualified=true means the current DEV execution paths satisfy the registered implementation/validity gates and are bound into the registered analysis. It is not evidence that A, B, G or P is scientifically true. Developmental scientific verdicts are not confirmatory evidence and the confirmatory namespace remains untouched.

Implementation qualification complete; final planning and text-implementation approval pending; study not frozen; no confirmatory evidence observed.